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Using Statistical Methods from Econophysics to Analyse Bubble Motion in a Two-Dimensional Foam

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ABSTRACT

A 2D foam coarsens over time, larger bubbles grow and smaller bubbles disappear. As this happens the foam restructures and the bubbles get displaced. This paper investigates the statistical properties of individual bubbles using tools borrowed from econophysics. Drawing a basis between bubble displacements and financial price movements, we apply log returns, volatility, temporal autocorrelation, complementary cumulative distribution functions (CCDFs), and mean squared displacement (MSD). Throughout this investigation, Brownian motion is used as a baseline, comparing with this pure randomness to characterise the particular type of randomness of these bubbles movements. We find that it is hard to draw conclusions from the bubble MSD so move on to more statistical methods to analyse. Bubble displacement distributions have a sharp peak around zero and taper off each side however for large Δt , looking at displacement over 10, 50, 100 timesteps or so, the distribution approaches a Gaussian curve. These distributions do not scale according to a Levy distribution like the financial distributions. The CCDF cut depends on where the cut is taken rather than being linear. A two parameter function of the form $f(C, \Delta t) = c_1(C)\Delta t / (1 + c_2(C)\Delta t)$ is proposed to describe the CCDF scaling. Notably, the autocorrelation function (ACF) of the log returns remains positive for the whole simulation, while the volatility ACF follows a power law decay similar to that of the financial data. These findings suggest that the bubble dynamics in a coarsening foam follow a particular non trivial statistical structure beyond simple Brownian motion.

ACKNOWLEDGEMENTS

I want to thank my supervisor, Professor Stefan Hutzler, for his guidance and support throughout this project. The weekly meetings made a huge difference in the progress made in this capstone even if it took me a while to programme things on the fly. Thank you Stefan.

I would also like to thank Professor Simon Cox for providing the foam data that made this project possible, and a thank you to Professor Peter Richmond for this insight into the econophysics side of the project.

I am grateful to Professor Cox and Professor Richmond for the insightful meetings throughout the year, which helped monitor progress and discuss future aims of this work.

A special thanks to my family and friends for their patience and support throughout this college year.

Finally I would like to address the use of Gen AI in this project. I used a combination of ChatGPT and Claude AI not for any conclusion on data, but to allow me to use intuition and creativity while programming without the drawback of lack of niche Python syntax. I used AI as a crutch to accelerate progress rather than replace intuition and critical thinking which I believe is the correct way to use in a project like this.

CONTENTS

1	Introduction	5
2	Background Theory	6
2.1	Brownian Motion	6
2.2	Econophysics	7
2.2.1	Log Returns	7
2.2.2	Volatility	8
2.2.3	Temporal Autocorrelation	8
2.3	Statistical Tools	9
2.3.1	Mean Squared Displacement	9
2.3.2	Displacement Distributions	10
2.3.3	Complementary Cumulative Distribution Function	10
3	Computational Methods	11
3.1	Organising the Data	11
3.2	Brownian Motion	13
3.3	Mean Squared Displacement	13
3.4	Log Returns	13
3.5	Displacement Distribution	14
3.6	Complementary Cumulative Distribution Functions	14
3.7	Temporal Autocorrelation Function	15
4	Results and Discussion	15
4.1	Bubble Paths	15
4.1.1	Average Area	17
4.1.2	Mean Squared Displacement	18
4.1.3	Displacement Distributions	18
4.1.4	Complementary Cumulative Distribution Functions Collapse	20
4.1.5	Autocorrelation Functions	23
5	Conclusion	24
6	Appendix A: Computational Method	26
7	Appendix B: Results	28

1 INTRODUCTION

Foams are among the most familiar yet physically complex systems encountered in everyday life. From the froth on your morning coffee to the structure of biological tissues, foams occupy a unique place in the study of soft matter. A foam is a layer of gas bubbles separated by a liquid medium, and the collective behaviour of these bubbles; how they coarsen over time and how the foam restructures because of this, is the core of what is being investigated in this report.

Coarsening is a fundamental property of these foams. It is the process by which bubbles exchange gas across their shared interfaces causing larger bubbles to get larger and smaller bubbles get smaller. Over time, the average bubble area increases and the number of bubbles decrease. During this coarsening, once the small bubbles disappear, the larger gain new neighbours and the foam experiences a restructure. Understanding the statistical character of the individual bubble trajectories, and the type of randomness they follow, within this evolving foam structure is the main motivation of this project.

In this project, we use data from a 2D foam simulated by Prof. Simon Cox. In this data we are given the positions of the bubble centres at each timestep, along with individual properties such as area and coordination number. Tools from econophysics are borrowed to investigate the dynamics and overall statistics of this foam. The central analogy is between the bubbles' displacements through the foam and the price movements in a financial market. In both cases, one is interested in the nature of the randomness involved. Whether it is completely random with simple Brownian diffusion, and whether there is a need to characterise a different type of randomness.

Brownian motion is used in this project as a baseline of randomness for comparison. It is well understood in its statistical properties; Gaussian displacement distributions, mean squared displacement that scales linearly in time, and vanishing temporal autocorrelation. This means we can use it as a sanity check for all of our methods, if we apply our methods to it and get the expected results, we can be more confident in our results when we apply the same methods to the bubble trajectories. Deviations from this baseline are of particular interest as they suggest the presence of underlying statistical properties of the foam that influence individual bubble motion.

The report is structured as follows: Section 2 reviews the statistical mechanics of Brownian

motion, the tools from econophysics used in this project; log-returns, volatility, temporal autocorrelation, etc., and how they can be intuitively applied to the bubble displacements. Section 3 describes the use of pandas to organise the data in a data-frame, extract information, and how computational methods are applied to this information. Section 4, presents the results and discuss their implications for understanding bubble motion in coarsening foams. Finally, section 5 summarises our findings and suggests directions for future research.

2 BACKGROUND THEORY

2.1 BROWNIAN MOTION

Brownian motion refers to the random movement of particles suspended in a fluid due to the cumulative effect of many independent collisions with the surrounding particles or molecules. This phenomenon was first observed by the botanist Robert Brown in 1827, who noticed pollen grains moving erratically in water.

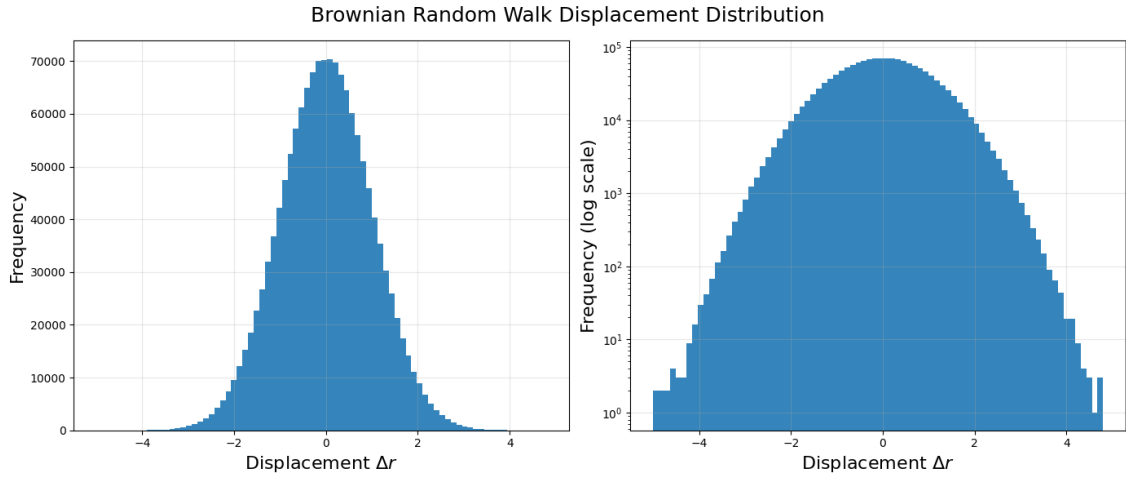


Figure 2.1: Averaging over 1000 Brownian random walks. As expected we get a Gaussian curve for the displacement distribution (left), and worth noting the inverted parabola on the log scale (right)

As mentioned in the introduction, since the statistical properties of Brownian motion are well known, we can use it as a sanity check to ensure our computational method is working accordingly. The displacements are picked from a Gaussian distribution, so are used as a reference point for all distributions. Since the displacements are random they are not correlated, in other words the autocorrelation function fluctuates around zero.

2.2 ECONOPHYSICS

In this section, I am going to first explain the application of these methods when analysing financial markets. Then I will explain how they intuitively can be applied to the bubble displacements to help us with our statistical analysis.

2.2.1 LOG RETURNS

Suppose $s(t)$ is the price of an asset at time t . One can measure the price fluctuations by simply looking at the change in price over a time δt . [4]

$$\delta s(t, \delta t) = s(t + \delta t) - s(t) \quad (2.1)$$

However, the main problem with looking at the price change in an asset is that it is indirectly affected by external economic events; recession, inflation etc. It is more useful to look at the **log returns**.

$$r(t, \delta t) = \ln \left(\frac{s(t + \delta t)}{s(t)} \right) = \ln(s(t + \delta t)) - \ln(s(t)) \quad (2.2)$$

The log returns are now seen to be additive.

$$r(t, \delta t_1 + \delta t_2) = r(t + \delta t_1, \delta t_2) + r(t, \delta t_1) \quad (2.3)$$

The incorporation of an inflation factor, $f(t)$, we get

$$\ln \left(\frac{f(t + \delta t)s(t + \delta t)}{f(t)s(t)} \right) = r(t, \delta t) + \ln \left(\frac{f(t + \delta t)}{f(t)} \right) \quad (2.4)$$

If we look at the second term, we see that if $f(t + \delta t) = f(t)$ then that term equals zero. The use of log returns takes account of inflation that remains constant over time. [4]

How this might apply to our bubbles is that we can use the log returns accountability of inflation to account for the overall growth of the bubbles in the foam over time as it coarsens. With this overall increase in bubble diameter the magnitude of the displacements increase aswell. For this reason, it is worth taking the log returns of the bubble displacements now instead so we can analyse the relative localised bubble dynamics rather than being affected by overall growth on top of that.

2.2.2 VOLATILITY

In a financial market, volatility is a measure of the magnitude of price fluctuations. High volatility markets, such as cryptocurrencies, are characterised by large price swings, while low volatility markets, such as government bonds, tend to have more stable prices.

We can define the volatility as the absolute value of the log returns [4]:

$$v(t, \delta t) = |r(t, \delta t)| \quad (2.5)$$

This can directly be applied to our bubble displacements, intuitively applied as a measure of how strong the fluctuations in the magnitude of the bubble displacements are. We can use this to investigate whether there are periods of high volatility in the bubble motion, and whether these periods are correlated with other properties of the foam such as the coordination number or area of the bubbles.

2.2.3 TEMPORAL AUTOCORRELATION

A temporal correlation is a measure of how a variable x at time t is influencing its value at a time $t + \tau$. An intuitive example is rolling across the room in a rolling chair. If I push off one wall while sitting in the chair, my position at that time is highly correlated with my position a second later, however that evening where I am writing my capstone thesis has nothing to do with whether I was rolling in a chair or not, in other words my position then is completely uncorrelated to my position when I rolled the chair.

This correlation measure is given by evaluating the *normalised autocorrelation function*, $R(\tau)$, defined as:

$$R(\tau) = \langle (x(t) - \langle x(t) \rangle)(x(t + \tau) - \langle x(t + \tau) \rangle) \rangle / (\sigma_t \sigma_{t+\tau}) \quad (2.6)$$

provided the standard deviation, $\sigma_t = \sqrt{\langle x^2(t) \rangle - \langle x(t) \rangle^2}$ is non-zero and finite [4].

Temporal autocorrelation can be applied in a financial market, the Dow Jones Industrial Average in this case. Figure 2.2 shows that the correlation drops quickly to zero and fluctuates around there in the daily return, however the autocorrelation function of the volatility follows a power law.

This can be directly applied to our bubble sample to see how long to the bubble displacements stay correlated for or if there is there any correlation at all after a topological restructure event.

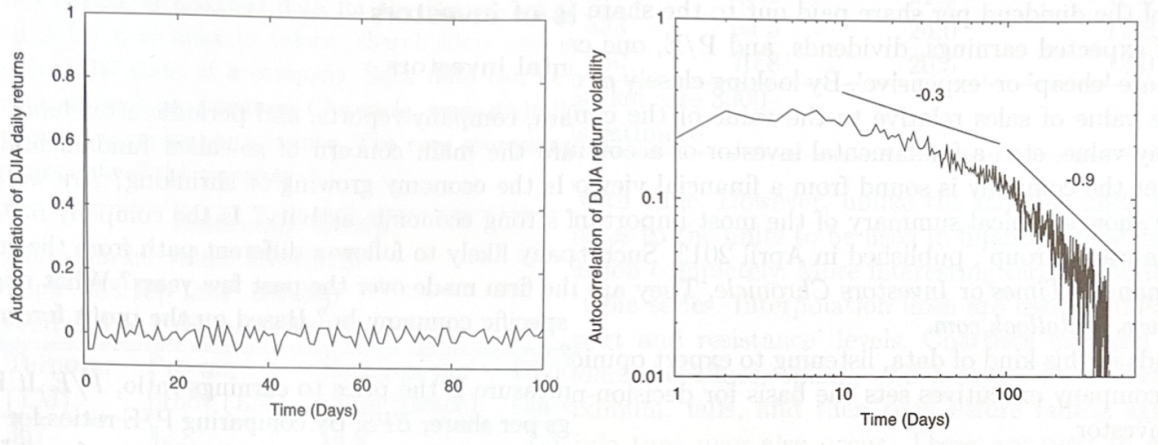


Figure 2.2: Dow Jones Industrial Average, autocorrelation of daily returns fluctuate around zero, autocorrelation of the volatility follows a power law. [4]

2.3 STATISTICAL TOOLS

2.3.1 MEAN SQUARED DISPLACEMENT

Mean Squared Displacement (MSD) is a measure of the average squared distance travelled over a given time interval. It is commonly used in physics and chemistry to study the motion of particles in a fluid or gas. The MSD is defined as:

$$\text{MSD}(\delta t) = \langle (\delta r(t, \delta t))^2 \rangle = \frac{1}{N} \sum_{i=1}^N (\delta r_i(t, \delta t))^2 \quad (2.7)$$

where $\delta r_i(t, \delta t)$ is the displacement of the i -th particle at time t over a time interval δt , and N is the total number of particles. The MSD can provide insight into the underlying dynamics of the system, such as whether the motion is diffusive, subdiffusive, or superdiffusive. For example, in a purely diffusive system, the MSD scales linearly with time ($\text{MSD} \propto \delta t$), while in a subdiffusive system, the MSD scales slower than linearly ($\text{MSD} \propto \delta t^\alpha$ with $\alpha < 1$), and in a superdiffusive system, the MSD scales faster than linearly ($\text{MSD} \propto \delta t^\alpha$ with $\alpha > 1$).

Rewrite Definition: following Wikipedia source, need a better source

For Brownian motion, as described by Einstein in his 1905 paper, the MSD scaled linearly with time following the diffusion law [1]:

$$\langle r^2(\tau) \rangle = 2dD\tau \quad (2.8)$$

Where d is the dimension, D is the diffusion coefficient, and τ is the elapsed time. It is in our interest to see if the bubble MSD follows this diffusion laws alongside Brownian Motion or if there it follows another trend.

2.3.2 DISPLACEMENT DISTRIBUTIONS

The displacement of a bubble over a time δt is given by:

$$\delta r(t, \delta t) = r(t + \delta t) - r(t) \quad (2.9)$$

Analysing these displacements in different ways will give insight into the underlying statistical properties of the bubble motion, comparing with Brownian distributions to characterise randomness for example. In this case, varying the time lag Δt to see how the distribution changes will reveal information about the statistics of the dynamics of these bubbles.

2.3.3 COMPLEMENTARY CUMULATIVE DISTRIBUTION FUNCTION

When investigating the log returns of a financial market, a key underlying statistical property of the probability density is that if we vary Δt the distribution scales according to a Lévy distribution.

Figure 2.3a is the probability distribution functions for the log returns over different Δt , all overlayed on one plot with the axes shifted for visual purposes. The complementary cumulative distribution function (CCDF), see figure 2.3b is obtained by integrating the probability distribution functions (PDF) for $|x| > x$ giving the probability the log returns are greater than a certain value for each Δt PDF curve.

A property of scaling according to a Lévy distribution is that if you take a cut at a certain probability on the CCDF curve, is that it is independent of where the cut is taken as seen in figure 2.3c where the cut, normalised to the $\Delta t = 1$ cut, increases linearly with Δt .

Finally, the slope of this linear plot can be used to scale the x axis of the original plot (2.3a) by plotting over the x axis, $x' = x/\Delta t^\beta$, resulting in the collapse of all PDFs onto one *master curve* (Figure 2.3d). A Lévy curve can be fit to the peak of this plot using a Lévy fit parameter of $\alpha = 1/\beta$

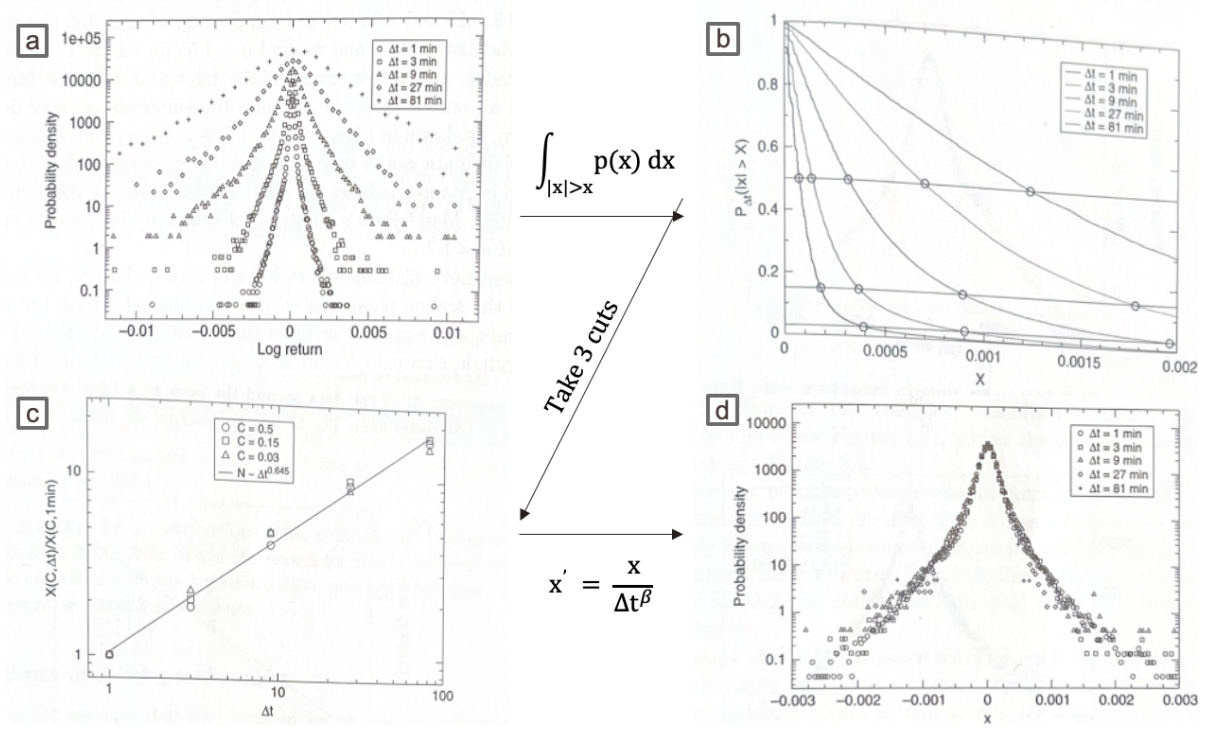


Figure 2.3: Series of figures which describes how complementary cumulative distribution functions can be use to collapse data onto a master plot. [4]

It would be interesting to apply to the bubble displacements and log returns to see if they also scale according to a Lévy distribution, if so, what would the scaling exponent be? If not, it is worth investigating what the underlying scaling properties are. It is worth noting, a key difference between these financial markets and the foam, is that as the foam coarsens the number of bubbles decrease over time, whereas there is constantly an abundance of data for the financial log returns.

3 COMPUTATIONAL METHODS

3.1 ORGANISING THE DATA

Simon's foam simulation data comes in the form of a csv (comma-separated value) file. To extract usable information for the python script, pandas is used to sort the data into a readable dataframe. *Sample of data... see Appendix A: Listing 2.*

```

1 df = pd.read_csv(
2     DATA_FILE,
3     comment="#",
4     sep=r"\s+",
5     header=None,
6     names=["id", "x", "y", "area", "pressure", "Z"]
7 )

```

Listing 1: Organising csv data file into a usable dataframe

Using a pandas dataframe allows easy sorting of the data from the python script. For example, when calculating log-returns, it is most efficient to sort by *id* first, then *timestep* to ensure correct ordering, and calculate the log-returns for that bubble as per the definition. Another example, when calculating the average area, we instead want to sort by timestep and use the *area* column to average over all bubbles in that timestep.

One of the first problems encountered when analysing the path of the bubbles was the **periodic boundary condition**. Figure 3.1 shows that the bubble paths that cross over the border jump across to the opposite side of the plot. This is how the data is presented in the text file.

To account for these jumps, using the length of the simulation box as found in the data file, a displacement limit of half the box can be used to flag large jumps, characterise whether it is in the x or y and rescale the x or y jump by the length of the box. Doing this leads to the proper paths of the bubbles as seen in figure 4.1.

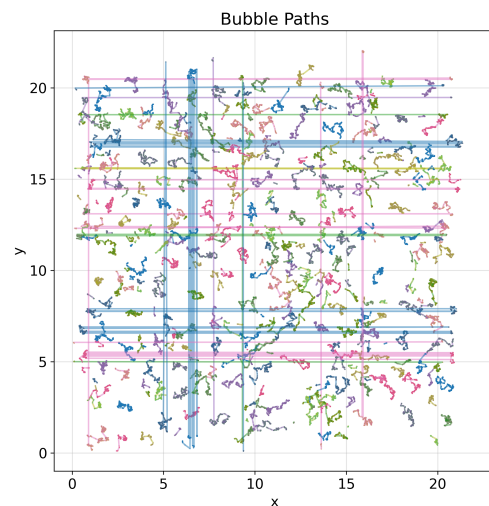


Figure 3.1: Periodic boundary conditions need to be accounted for.

The best way found to do all this in conjunction with the analysis was to have a `data_loader.py` file, where the data frame is made, the bubble paths are corrected and data files can be easily interchanged. All stored variables and data frames are then loaded into separate Jupyter notebooks for each separate statistical method to be applied and analysed.

3.2 BROWNIAN MOTION

In the interest of comparing directly with to the bubble motion, the Brownian walk parameters are chosen to match the dataset as closely as possible: 400 independent walks are made (equal the initial bubble count), each using 726 steps (equal the number of simulation timesteps), with step sizes taken from a zero-mean normal distribution. The choice of standard deviation is trivial as it is the shape of the statistics, displacement distributions, correlation functions, etc., that we are concerned with. For the MSD, the $2Dt$ relation can be plot on dimensionless axes to be independent on Brownian simulation parameters.

3.3 MEAN SQUARED DISPLACEMENT

To calculate the MSD, for each timestep, the squared displacement of each bubble (or Brownian step), $\delta r^2 = \delta x^2 + \delta y^2$, is calculated. As the name implies, the mean of the squared displacement is calculated and stored. Once this is done for all timesteps, the MSD can be plot versus time visualise the data.

An important difference when investigating the bubble MSD is that the number of bubbles is changing as seen in the survival curve, *see Appendix B: 7.2*. Noting the milestones (300, 200, 100 bubbles for example), while plotting the MSD will be helpful in coming to a conclusions on how much of an effect the sample size has on the MSD relation, as seen in figure 4.4.

3.4 LOG RETURNS

Calculating the log returns is as simple as following the definition, equation 2.2, using x and y position as an analogue to the price $s(t)$. It is important to note that the radial position is used $r(t) = \sqrt{x(t)^2 + y(t)^2}$ since it is easier to analyse the autocorrelation function later, given concatenating the x log returns and y log returns (or displacements for that matter) wouldn't make physical sense for the ACF and volatility as they have temporal dependence (as oppose to a distribution where it is the magnitude of interest).

3.5 DISPLACEMENT DISTRIBUTION

The displacement distribution is trivially calculated by looping over bubble id, sorting by timestep, and using the definition of displacement $\delta x(t, \Delta t) = x(t + \Delta t) - x(t)$ to calculate displacement in the x and y separately. For the distributions, looking at figure 3.2 it seems sufficient to concatenate the x and y displacements to get double the dataset and better overall statistics. However as mentioned in the previous section, the radial displacement will need to be taken into account instead when looking at the autocorrelation function. Varying Δt can be done by looping over a list of Δt 's to analyse displacements over different time lags.

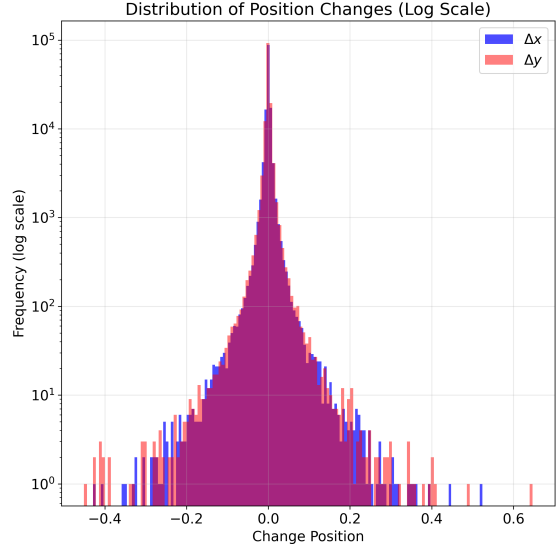


Figure 3.2: x and y displacement distribution line up well enough to combine and get double the data.

3.6 COMPLEMENTARY CUMULATIVE DISTRIBUTION FUNCTIONS

Using the CCDF to collapse the distributions onto a master plot consists of four main steps. The first, simply overlaying the displacement distributions on the one plot for varying Δt and shifting the data for visual purposes so the distributions do not overlap. Second, using an integration method to obtain the CCDF itself. Third, taking probability cuts in the CCDFs (storing particular values in a list while calculating the CCDF) and plotting these values versus Δt to get the scaling relation. Finally, using the slope of this CCDF scaling function to rescale the x axis of the original plot, using $x' = x/\Delta t^\beta$ where β is the slope, to collapse all distributions onto one master plot... in theory.

The numerical method used to find the CCDF is a sort-and-count method (`np.searchsorted`), which gives $P(|x| > X) = 1 - F(X)$ where $F(X)$ is the fraction of samples $\leq X$. This is a standard non-parametric Empirical Cumulative Distribution Function estimator, equivalent in principle to `matplotlib.axes.ecdf`, but evaluated over a uniform grid to allow direct comparison across Δt 's. This method is exact with respect to the sample does not introduce discretisation error unlike histogram-based approaches which require binning and therefore

only approximate the ECDF [2]. Methods such as Simpson’s Rule are similarly unsuitable, as they require an estimation of the PDF, introducing approximation error before integration even begins.

If the slope β is taken from figure 4.8a, then $1/\beta$ is used as the Lévy parameter fitted to the peak of the distribution. When investigating the scaling properties of the CCDF further as seen in figure 4.1.4, the guess function (equation 4.1) is fit trivially via a least square fit to find the free parameters c_1 and c_2 associated with the fit.

3.7 TEMPORAL AUTOCORRELATION FUNCTION

As mentioned before, it is important to ensure the displacement and log returns in the x and y are calculated radially and not concatenated as there is a time dependence, and if the y data only kicks in halfway through the analysis of the ACF then we lose physical meaning.

In this project, the *normalised autocorrelation function* is used, defined in section 2.2.3. For each lag τ , the ACF is computed directly as per it’s definition equation 2.6, *as seen in Appendix A: Listing 3*.

4 RESULTS AND DISCUSSION

4.1 BUBBLE PATHS

Figure 4.1 shows the path of the centres of the bubbles during the simulation. The mean position of the bubbles is shown in the centre of the plot to investigate for an overall drift in the bubbles. Mainly, this is to prove that there is no bias in displacements in the x and y direction, so when we look at the distribution of displacements, we can combine the x and y displacements for double the data.

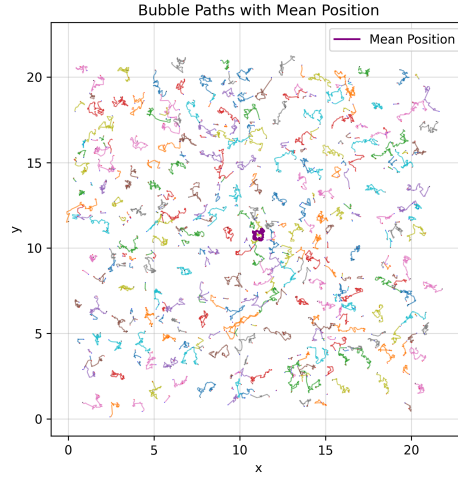


Figure 4.1: Paths of the centres of the bubbles.

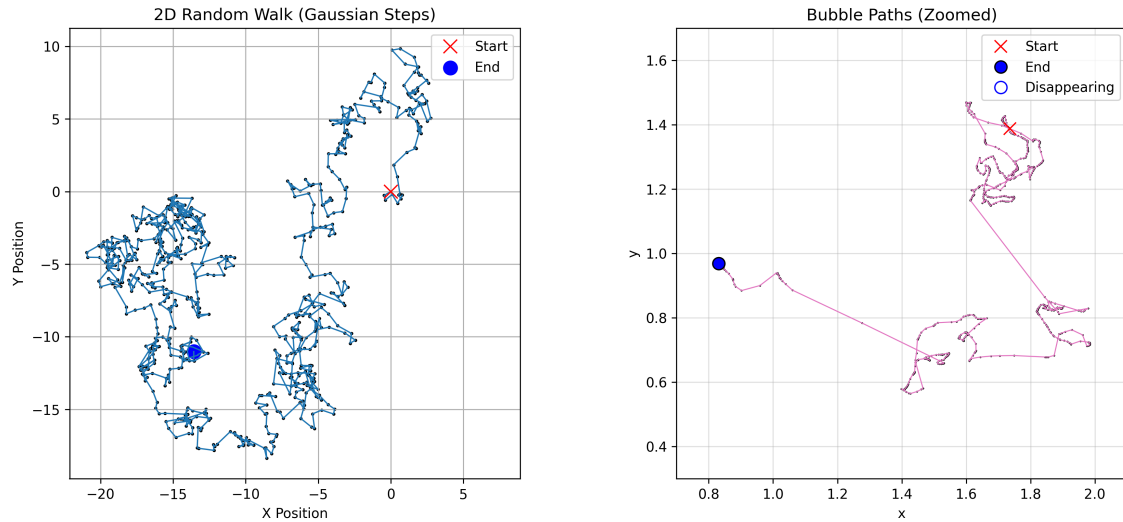


Figure 4.2: Comparison of a Brownian random walk (left) and a tracked bubble path (right) over the same number of timesteps.

Figure 4.2a shows a Brownian random walk over 726 steps, the same number of time steps in the foam simulation. Figure 4.2b shows the path of one of the bubbles which survives the full 726 timesteps. Comparing the two side by side, it is clear that the Brownian random walk is completely random, erratic, and uncorrelated. Whereas the bubble path seems slightly more structured, with some steps clustered together, when the foam is relaxed, and some large jumps

then the foam is restructuring. This is a stark difference between the two, however it is worth comparing with the Brownian walk as a baseline to see how random, or what particular type of randomness, the bubble paths follow.

4.1.1 AVERAGE AREA

As a sanity check, the average area over time is shown to be linear in figure 4.3. This was proven in two ways. The first was using the raw diameter data and averaging that the area obtained from that over all the bubbles in that timestep, The second method used an approximation, $\langle A \rangle = A_{box}(1 - \phi)/N(t)$, where ϕ is the liquid fraction and $N(t)$ is the number of bubbles at time t . Figure 4.3 proves these two methods line up exactly.

The key takeaway from this is that since the area is increasing linearly with time then we are losing bubbles while this is happening.

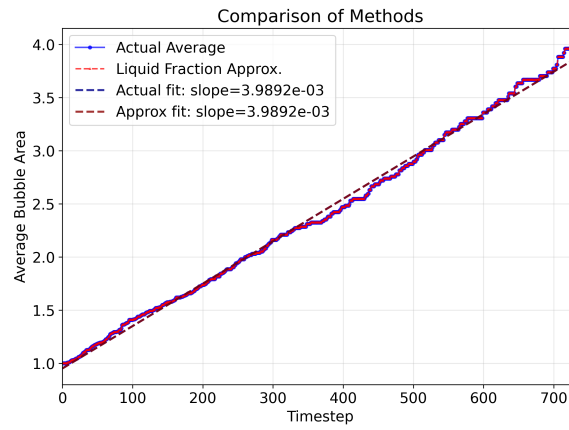


Figure 4.3: Average bubble area increases linearly over time.

4.1.2 MEAN SQUARED DISPLACEMENT

The mean squared displacement (MSD) was averaged over 400 Brownian random walks, and as expected scaled linearly with time. In figure 4.4, this Brownian MSD line is superimposed onto the plot of the bubbles MSD over time. There are some stark differences. Initially, since I was looking for this linear relation in the bubbles I tried to convince myself there was some sort of linearity in the plot. While there certainly is an upward trend, it is definitely not linear, at least not with the amount of data we have.

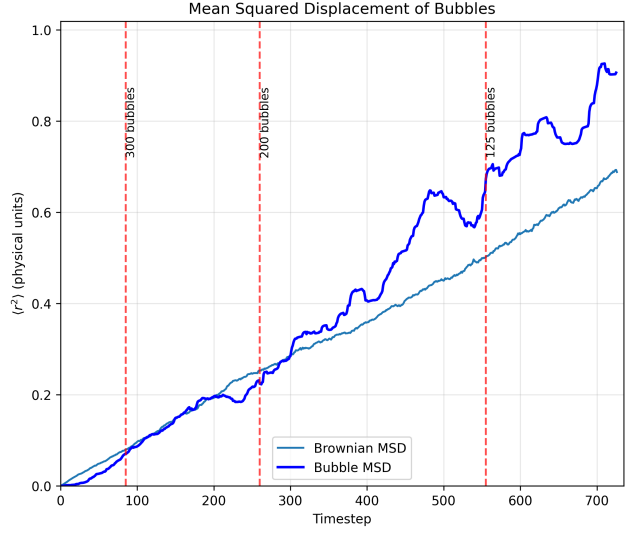


Figure 4.4: Hard to draw conclusion from bubble MSD

One clear distinction between the two simulations is that the number of bubbles decrease over time. This can be seen in the figure 4.4 with the red dashed lines marking the milestones in the bubble counts. As the number of bubbles decrease, the statistics are average over fewer paths, leading to increased noise making it harder to determine a linear relation. An example of this is plotting the Brownian MSD averaged over 100 walks imposed on a plot of MSD averaged over 400 walks (Appendix B: Figure 7.1). Averaging over a smaller number of trajectories leads to a noticable increase in noise. When analysing the bubble MSD, this can be seen once half the bubbles have disappeared around the 260th timestep. Having said that, even when we have the most amount of bubbles, 400-300 bubbles between $t=0$ and $t=80$, this section of the plot almost looks like it follows a power law so it is hard to draw conclusions from this investigation.

The conclusion from this investigation is that it is hard to make any good conclusions from this analysis. For that reason, we move on to other method to analyse the statistics of this system.

4.1.3 DISPLACEMENT DISTRIBUTIONS

The displacement distribution of the bubble displacements looking at every timestep ($\Delta t = 1$) has a sharp peak around 0 and tapers down each side, as seen in figure 4.5 on a log-lin scale. Comparing with the Brownian displacement curve in figure 2.1, there is a stark difference be-

tween the two distributions; an inverted parabola for the Brownian motion, and a sharp peak for the bubble displacements.

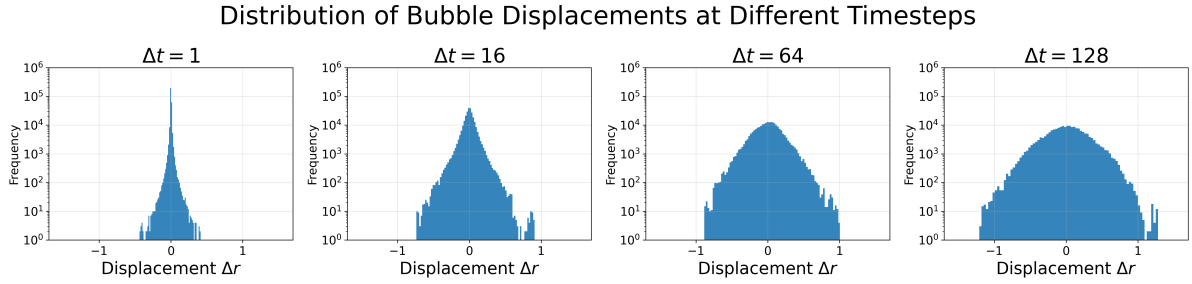


Figure 4.5: Displacement distributions looking at different time lags. Distributions widen towards a Gaussian distribution at larger time lags.

As Δt increases the bubble displacement distributions widen, eventually approaching a Gaussian shaped inverted parabola. This shows the uncorrelation comparing displacement at a certain timestep versus displacement a hundred or so timesteps later. There is an element of randomness which brings the Gaussian distribution into play.

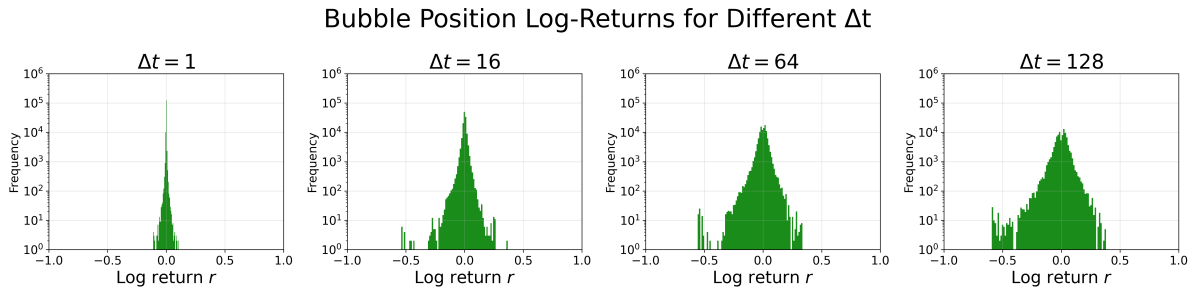


Figure 4.6: Log return distributions over increasing Δt . Distribution does not widen into a Gaussian unlike the displacement distributions. Log-linear scale.

Investigating the log return distributions led to a different conclusion than the displacement distributions. Unlike the displacement distributions, the log return distribution does not widen into a Gaussian curve. As the distribution don't evolve into a Gaussian, section 4.1.5 will investigate if there is an element of time correlation in the log returns.

4.1.4 COMPLEMENTARY CUMULATIVE DISTRIBUTION FUNCTIONS COLLAPSE

As confirmation of the CCDF collapse method used in this project, the Brownian walk distributions (both displacement and log returns) collapse as intended onto one master curve, *see Appendix B: Figures 7.3 - 7.6*. One observation in particular is that the CCDF cuts scale exactly linearly with the time lag Δt with a slope of 0.5. This was to be expected as the displacements for the Brownian walk are chosen from a Gaussian distribution with standard deviation which scales as $\sigma(\Delta t) \sim \sqrt{\Delta t}$ [3]. Since the entire distribution rescales with $\sigma(\Delta t)$, every quantile, including $X(C, \Delta t) \sim \sigma(\Delta t) \sim \Delta t^{0.5}$.

Displacement distributions from the section 4.1.3 were used for an initial attempt at a Complementary Cumulative Distribution Function (CCDF) collapse.

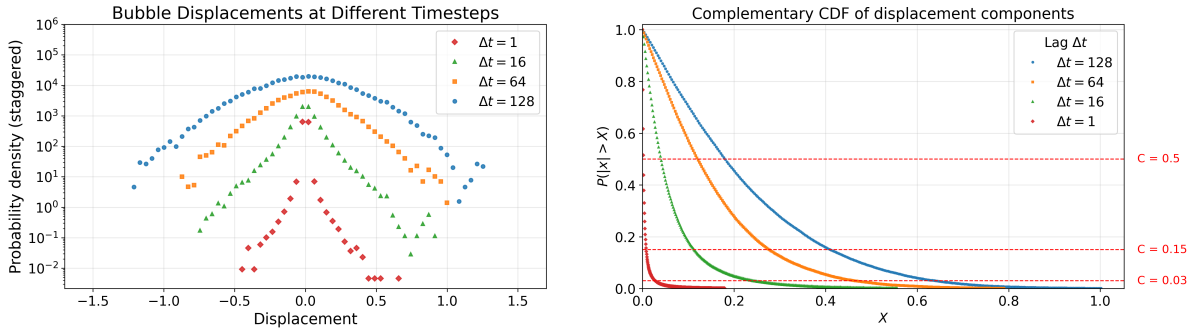


Figure 4.7: Displacement distributions overlayed on the one plot, axes shifted for visual purposes (left). These PDFs were then integrated over $|x| > x$ to get the CCDF (right).

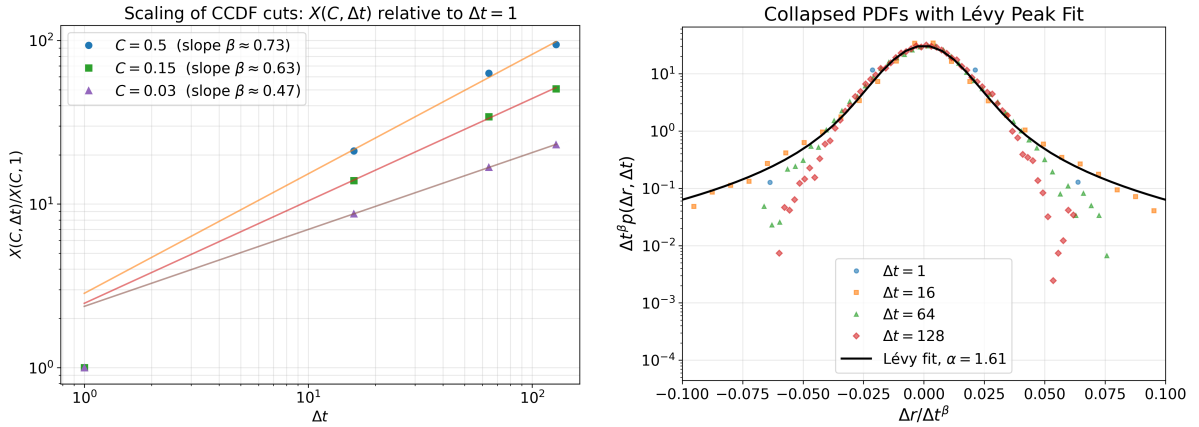


Figure 4.8: Bubble displacements do not scale according to Levy distribution.

A key difference between the bubble displacement distribution scaling and the scaling of financial markets is that the CCDF cuts are dependent on where the cut is taken, there is not a linear relation between the cut (normalised to the first cut) versus Δt , *see figure 4.8a*. For arguments sake, the middle slope is taken, $\beta = 0.63$ giving a relatively good collapse in figure 4.8. The centre of the distribution collapses well, however, the tails clearly deviate from this scaling a lot.

The scaling of the log returns follows the same trajectory as the displacement distributions. The collapse works relatively well for an average CCDF scaling slope. However, collapse breaks down in the tails quickly. *See Appendix B: Figures 7.7, 7.8.*

To investigate this scaling further, more Δt distributions were added along with more CCDF cuts in an attempt to fit a function to these scaling properties. (*See Appendix B: figure 7.9*). Figure 4.9 shows the resulting scaling of each cut with Δt .

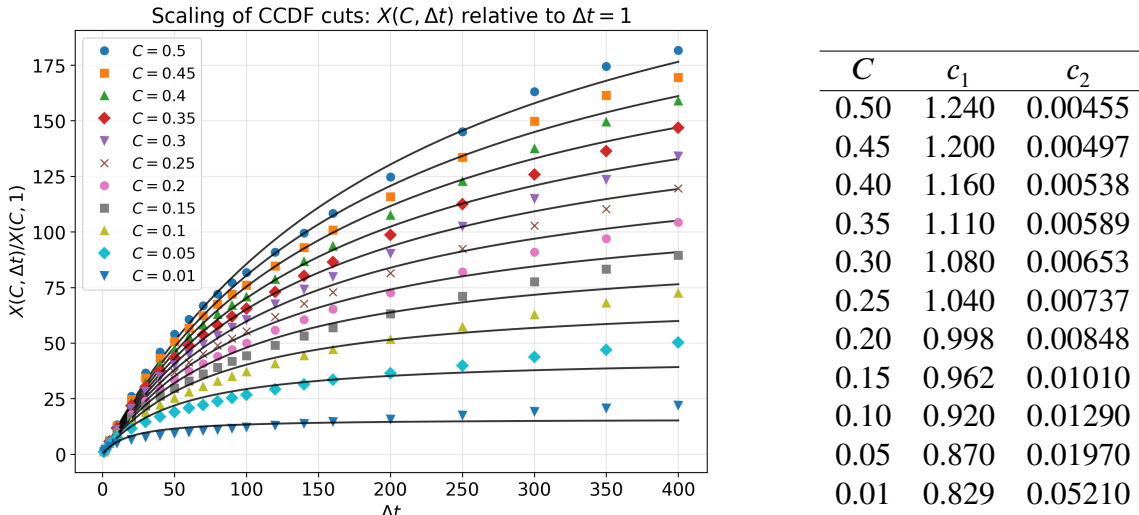


Figure 4.9: Bubble displacements do not scale according to Levy distribution (left). Fitted parameters for each CCDF cut (right).

By process of trial and error to find a function which scales this way, equation 4.1 seemed the best fit.

$$f(\Delta t) = \frac{c_1 \Delta t}{1 + c_2 \Delta t} \quad (4.1)$$

Equation 4.1 uses two free parameters, c_1 and c_2 , and the fit uses a least square method to obtain the corresponding c_1 and c_2 for the best fit. Each free parameter value is stored alongside the

cut C that these values are associated with. Plotting c_1 against C gives a linear relation and c_2 versus C gives a decay power law which was estimated using a non-linear least squares fit as shown in figure 4.10.

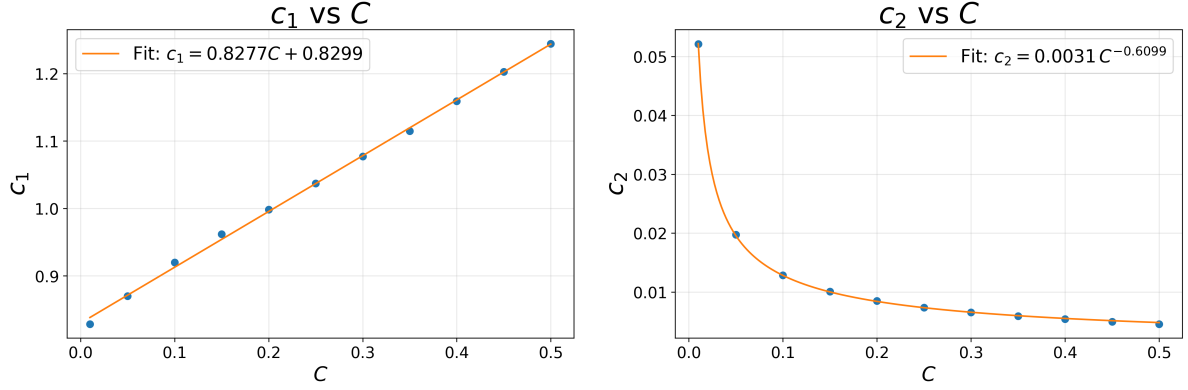


Figure 4.10: Linear relation for scaling of c_1 with respect to C , and power law fit of c_2 vs C

Using the plots and best fits in figure 4.10, a closed form for c_1 and c_2 has been approximated:

$$c_1 = 0.83(C + 1) \quad (4.2)$$

$$c_2 = 0.0031 C^{-0.61} \quad (4.3)$$

given more time I am hoping to somehow use this slope and $C = 0$ value to scale the displacement distribution PDFs onto one masterplot.

Peak scaling method, appendix maybe

4.1.5 AUTOCORRELATION FUNCTIONS

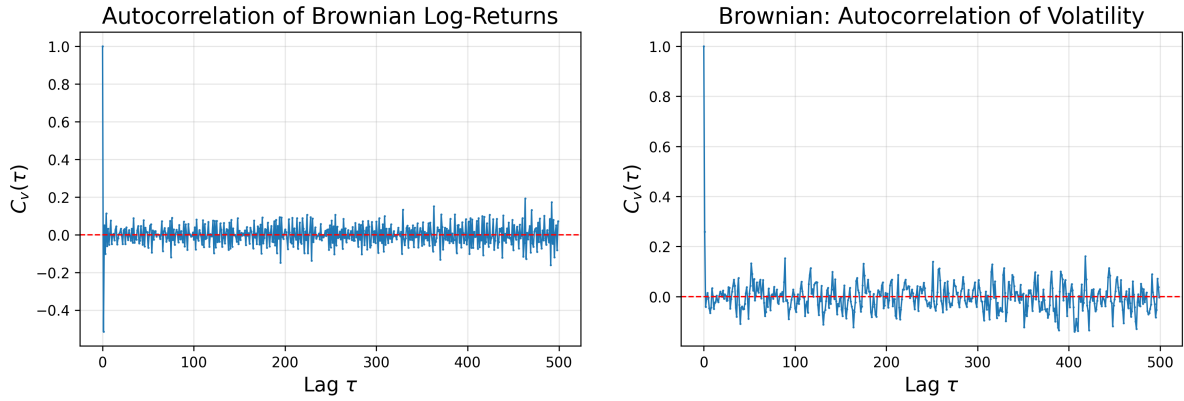


Figure 4.11: There is no correlation in Brownian motion, log returns or volatility in this case, as it is the baseline example of pure randomness

As expected, the Brownian log return and volatility autocorrelation functions drop straight to zero and fluctuate there for the rest of the simulation.

Figure 4.12 shows the autocorrelation function for the displacement of the bubbles, as expected there is no correlation. More interestingly, the autocorrelation function for the log returns is always positive, as seen in figure 4.13 (left). It is hard to conclude the exact physical intuition behind this but it is a key finding in this investigation nonetheless as it is unexpected that it is correlated like this.

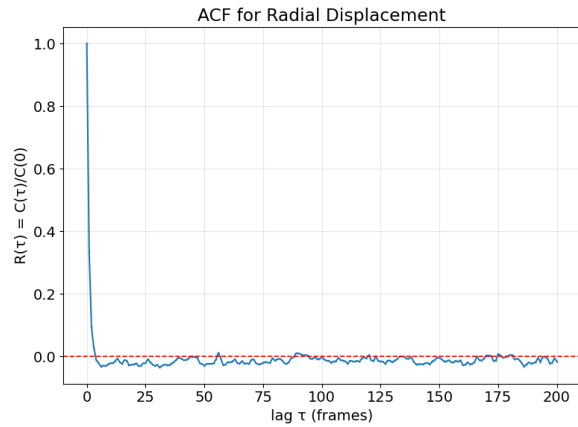


Figure 4.12: Bubble displacements remain relatively uncorrelated, ever so slightly negatively correlated

Figure 4.13 (right) shows the autocorrelation of the volatility. The trend follows a power law of $\tau^{-0.4}$ and quickly drops to zero towards the end. This can be related back to figure 2.2. The volatility autocorrelation function seems to follow a similar sort of trend, however where the DJIA differs is that the daily returns fluctuate around zero whereas the log return autocorrelation function is always slightly positive.

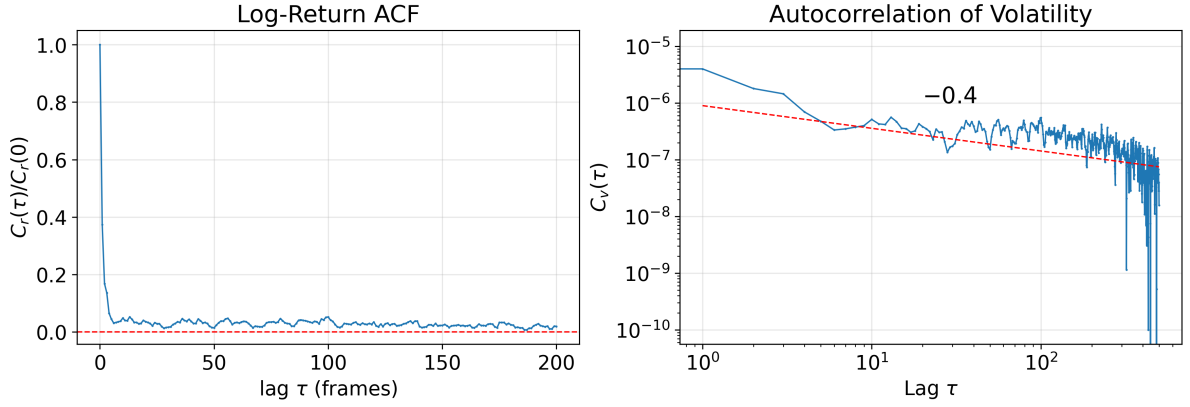


Figure 4.13: Log return ACF unexpectedly stays ever so slightly positive, the volatility ACF follows a power law trend that can be linked back to the DJIA figure 2.2.

5 CONCLUSION

In this project, we applied statistical tools from econophysics to analyse the motion of bubbles in a coarsening two-dimensional foam, using Brownian motion as a baseline of pure randomness to determine the particular type of randomness of the bubble dynamics.

The bubble paths were compared with a Brownian walk of the same amount of time steps. While the Brownian walk looks erratic and uncorrelated, the bubble path looks to have more structured motion. Small clustered steps when the foam is relaxed, and large correlated jumps when the foam is restructuring. This inspired further investigation into the underlying statistics.

The mean squared displacement for Brownian walks follows the linear relation $\langle r^2(\tau) \rangle = 2D\tau$. The MSD for the bubbles were analysed in hopes to find the same relation. While the bubble MSD seems to follow an upward trend, the more bubbles disappear, the less data we have to average over, so it deviates a lot. It was hard to draw conclusions from this so we moved on to other statistical methods.

The displacement distribution deviate significantly from the Gaussian distribution of the Brownian motion, a sharp peak around the centre that tapers off. However, when we look at larger time lags Δt , the displacement distribution turns into a Gaussian distribution as the displacements become essentially random. More interestingly, for larger Δt in the log returns distribution the central peak seems to persist, hinting at an underlying correlation in the log returns.

The CCDF analysis showed that the bubble distributions do not scale according to a Lévy distribution, unlike the scaling of the financial log returns. Instead of being linear, it was found that the function with two free parameters, $f(C, \Delta t) = c_1(C)\Delta t/(1 + c_2(C)\Delta t)$, describes the scaling of the CCDF cuts pretty well. A best fit was used on c_1 vs C and c_2 vs C , to get a closed form; $c_1 = 0.83(C + 1)$, $c_2 = 0.0031 C^{-0.61}$. Using these to obtain a fully collapsed master plot remains an goal for future work.

The analysis of the autocorrelation functions showed some interesting results. While the bubble displacements showed no significant correlation, as expected, the log returns ACF drops close to zero quickly however remains positive for the entire simulation. The volatility ACF, following the behaviour of the Dow Jones Industrial Average volatility ACF, follows a power decay law of $\tau^{-0.4}$, suggesting the bubble volatility share some statistical structure with financial markets.

Future work could explore the scaling of the CCDF further. A closed relation of the free parameters gives hope that they could be used in some way to collapse the cut scaling curve and use a feature of this collapsed plot such as the slope to collapse the original distributions onto a masterplot. Extending the analysis of the foams to different foam samples of different lengths, numbers of bubbles, and liquid fractions would also help to establish how general our findings are.

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- [1] A. Einstein. *Über die von der molekularkinetischen Theorie der Wärme geforderte Bewegung von in ruhenden Flüssigkeiten suspendierten Teilchen*. Annalen der Physik (in German), 1905.
- [2] Matplotlib Development Team. *Cumulative Distributions — Matplotlib Documentation*. 2024. URL: https://matplotlib.org/stable/gallery/statistics/histogram_cumulative.html.
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6 APPENDIX A: COMPUTATIONAL METHOD

DATA SAMPLE

```
1  #400 bubbles
2  #width=20.203051, height=20.203051, time =1
3  ###vcentrex= 10.039337, vcentrey= 10.434378, vcentreAx= 10.000768,
vcentreAy= 10.350760, bcentrex= 10.584868, bcentrey= 10.863349,
bcentreAx= 10.790131, bcentreAy= 10.862649
4  #id,,x,y,area,pressure,Z
5  1 16.830467 10.024328 1.505199 1.051348 16
6  2 10.836769 7.222546 0.273170 2.395116 5
7  3 8.349003 1.681417 1.030034 1.177891 9
8  4 1.364649 6.059485 0.298862 2.014220 5
9  5 1.334989 3.429860 0.151537 3.716496 4
10 6 9.045682 19.629107 0.385979 2.093547 5
11 ...
```

Listing 2: Sample of foam simulation data (first 6 of 400 bubbles for the first timestep)

AUTOCORRELATION CODE

```
1 def scalar_acf_direct_nonstationary(x, max_lag=None):
2     x = np.asarray(x, dtype=float)
3     n = len(x)
4     if max_lag is None or max_lag > n - 1:
5         max_lag = n - 1
6
7     x = x - x.mean()
8
9     acf = np.empty(max_lag + 1, dtype=float)
10    for tau in range(max_lag + 1):
11        x_t = x[:n - tau]
12        x_t_tau = x[tau:]
13
14        sigma_t = np.std(x_t)
15        sigma_t_tau = np.std(x_t_tau)
```

```
16
17     denom = sigma_t * sigma_t_tau
18     if denom == 0:
19         acf[tau] = 0.0
20     else:
21         acf[tau] = np.mean(x_t * x_t_tau) / denom
22     return acf
```

Listing 3: Calculating the normalised autocorrelation function

7 APPENDIX B: RESULTS

MEAN SQUARED DISPLACEMENT

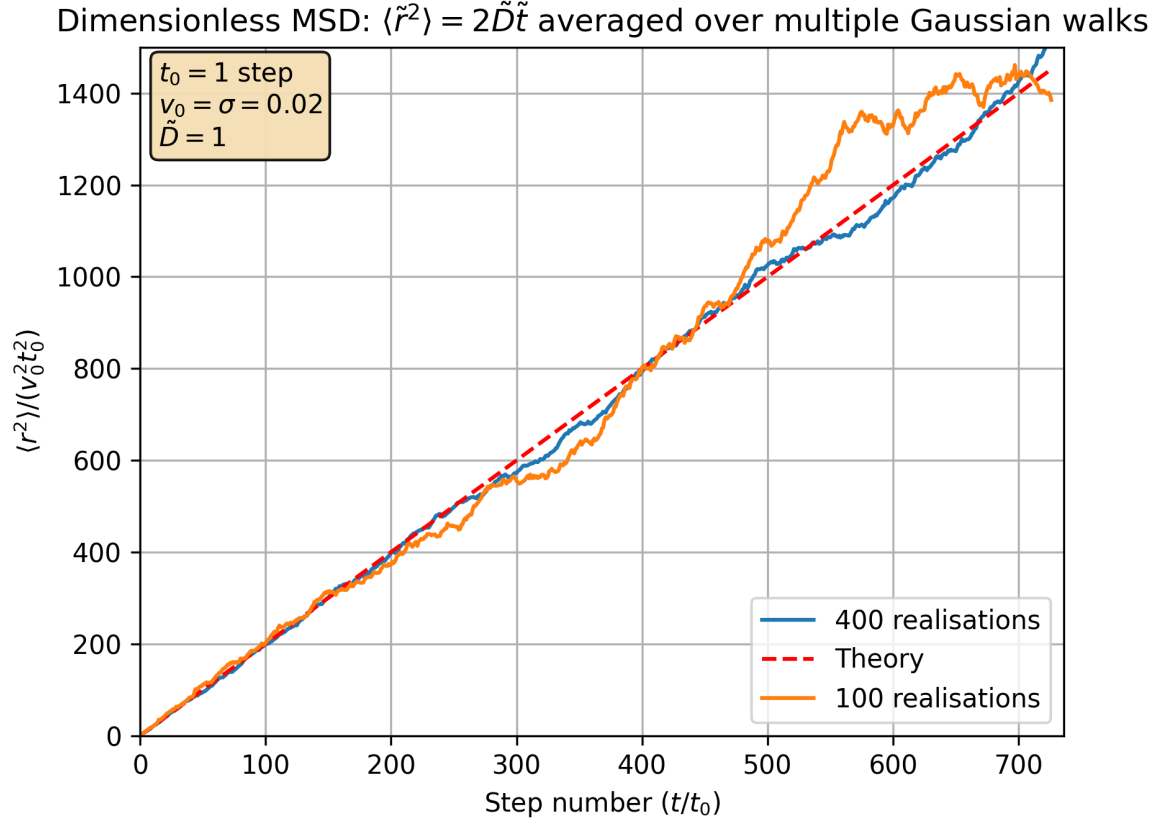


Figure 7.1: MSD for Brownian walks. Dimensionless axes allow the plot to be independent of simulation parameters and give a slope of 2 as per the diffusion law $\langle r^2 \rangle = 2Dt$. Averaging over both 100 and 400 walks to show the effect of lack of sample size on the statistics.

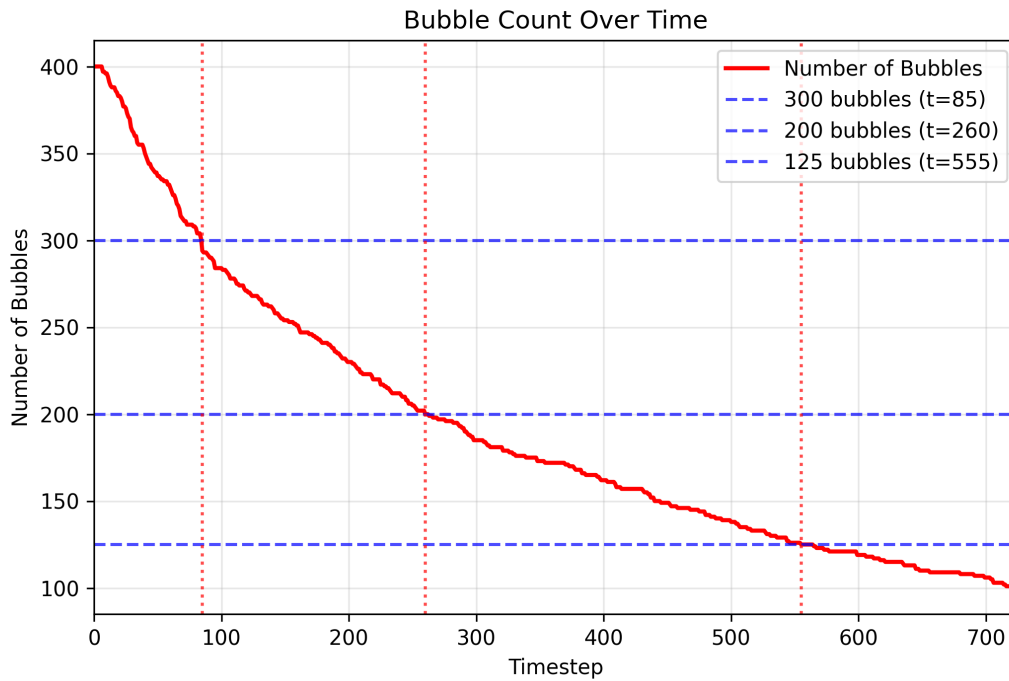


Figure 7.2: Survival curve of bubbles over the duration of the simulation with milestones marked.

CCDF SCALING

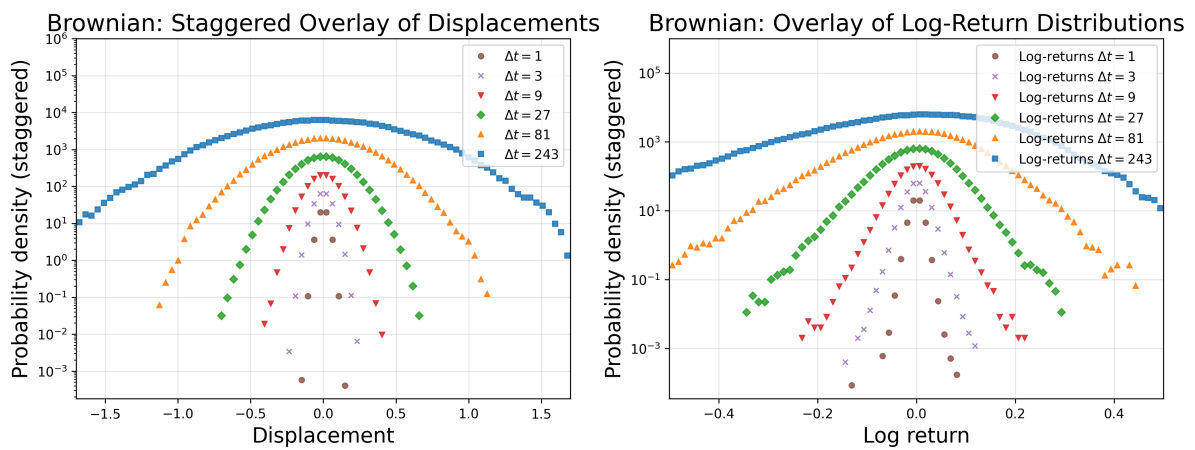


Figure 7.3: Brownian Distributions Overlaid

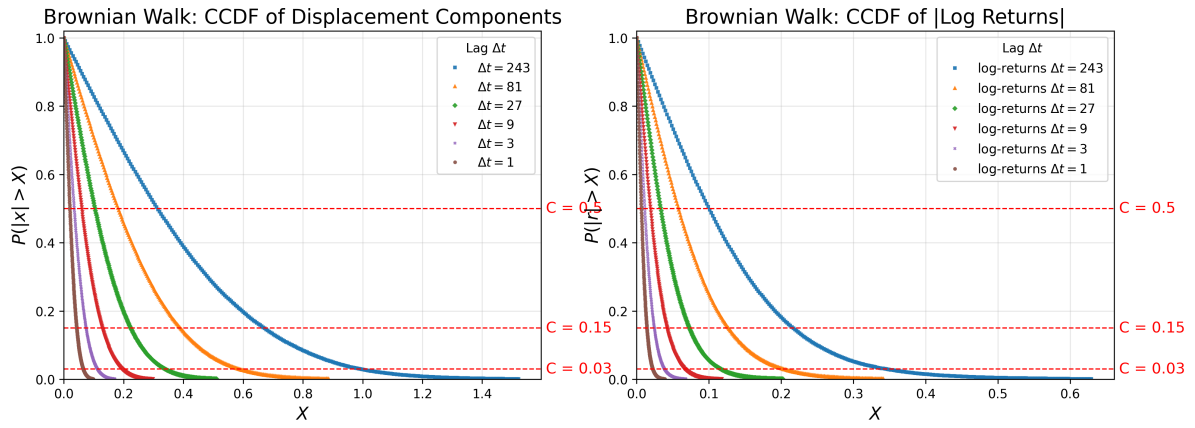


Figure 7.4: CCDF for Brownian Displacement Distributions and Log>Returns Distributions

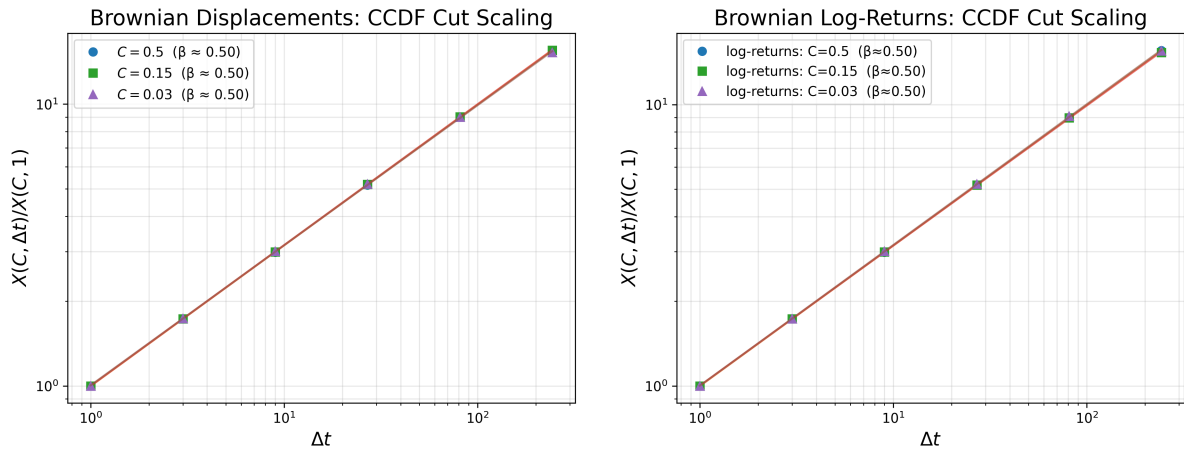


Figure 7.5: CCDF scaling for Brownian, both displacements and log returns scale as expected with an exponent of $\beta = 0.5$

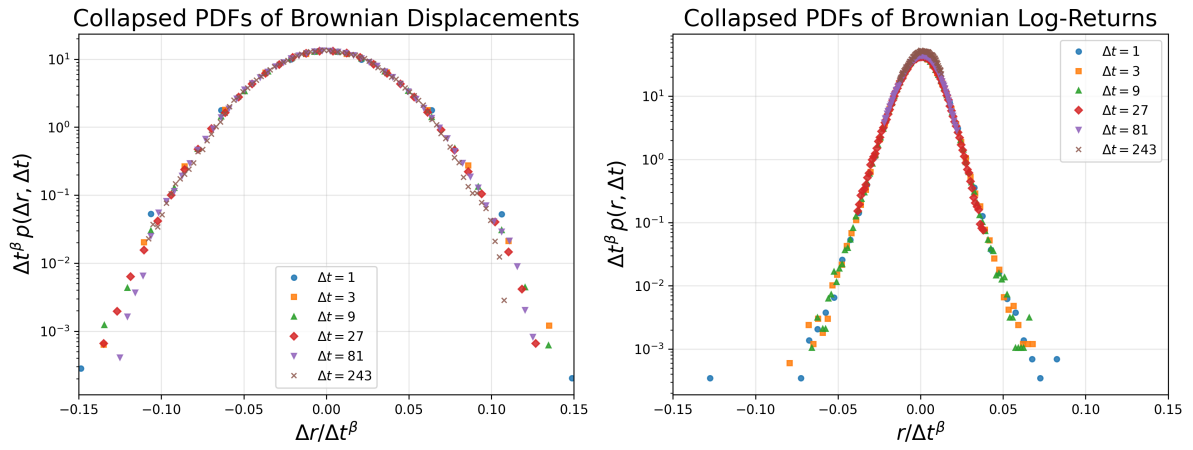


Figure 7.6: Using $\beta = 0.5$ slope to scale the x axis of the original distribution plots, all distributions collapse onto the one master plot as intended

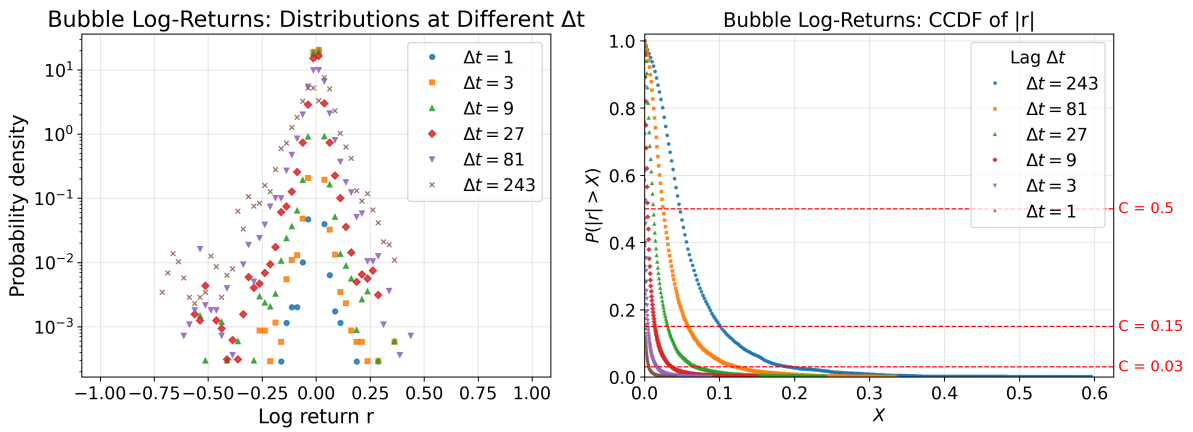


Figure 7.7: Log return distributions and CCDF curve as procedure

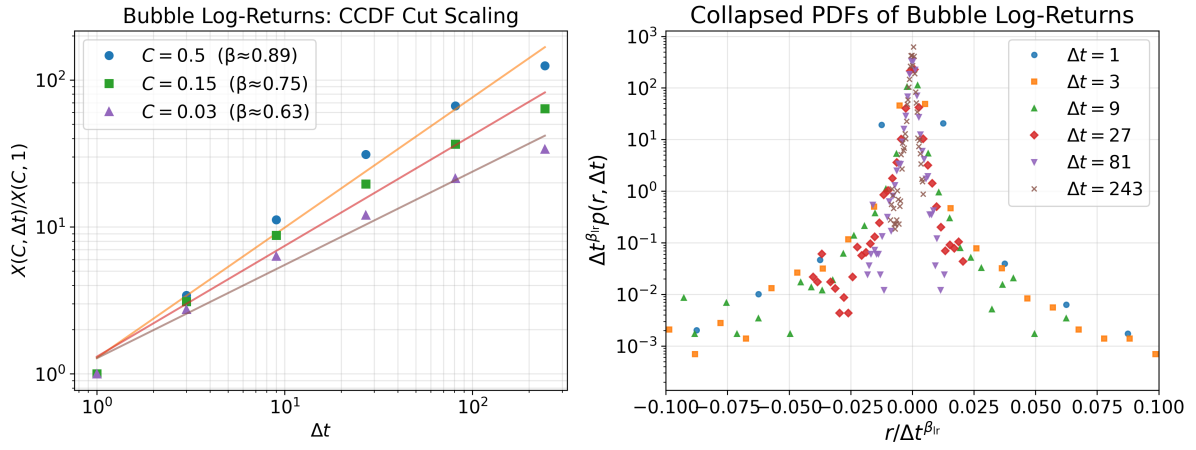


Figure 7.8: Log returns follow the same CCDF scaling as the displacements

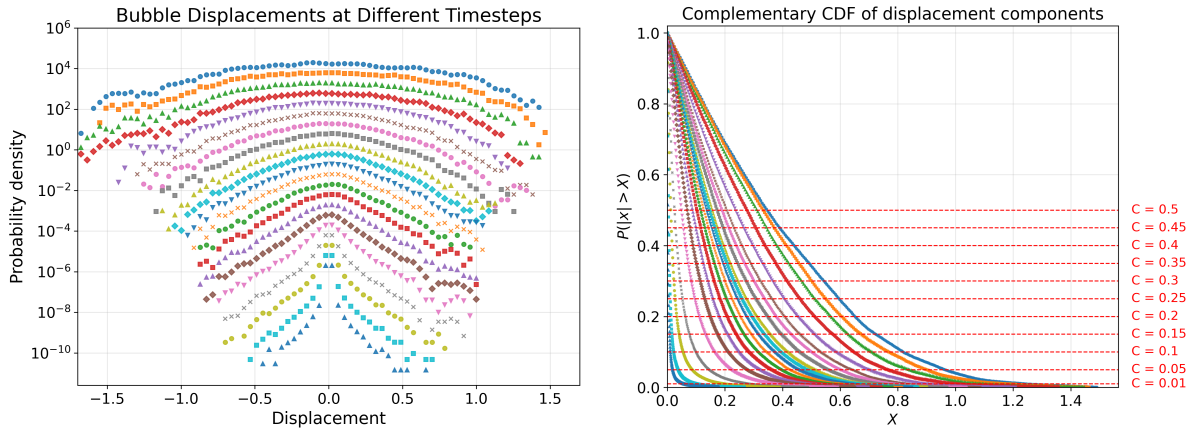


Figure 7.9: Bubble displacements do not scale according to Levy distribution.